

Systems Science & Control Engineering

An Open Access Journal

ISSN: 2164-2583 (Online) Journal homepage: www.tandfonline.com/journals/tssc20

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**To cite this article:** Muhammad Muzamil Aslam, Ali Tufail, Liyanage Chandratilak De Silva, Rosyzie Anna Awg Haji Mohd Apong & Abdallah Namoun (2024) An improved autoencoder-based approach for anomaly detection in industrial control systems, Systems Science & Control Engineering, 12:1, 2334303, DOI: 10.1080/21642583.2024.2334303

**To link to this article:** https://doi.org/10.1080/21642583.2024.2334303

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Published online: 18 Apr 2024.

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SYSTEMS SCIENCE & CONTROL ENGINEERING: AN OPEN ACCESS JOURNAL
2024, VOL. 12, NO. 1, 2334303
https://doi.org/10.1080/21642583.2024.2334303

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# An improved autoencoder-based approach for anomaly detection in industrial control systems

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**ABSTRACT**

Security is a complex issue in critical infrastructure like industrial control systems (ICS) since its leakages cause critical damage. Protecting the ICS environment from external threats, cyber-attacks, and natural disasters is important. Various works have been done on anomaly detection in ICS, and it has been identified that these proposed approaches are the sole models associated with them. Although there is a research gap in anomaly detection methodologies because of their computational complexity. To overcome the research gap of high false positive rate (precision), accuracy, and computational complexity in the literature, the study presents an Improved autoencoder (ImpAE) anomaly detection methodology for anomaly detection in ICS. The proposed methodology is a deep learning-based model to build anomaly detectors that alarm the attacks affecting ICS security. This methodology follows a flexible and modular design that permits a group of numerous detectors to get suitable detection. To express the suitability of the proposed model, we implemented it on the Secure water testbed (SWat) dataset, which is collected from a working water treatment plant. Experimental work shows that by using ImpAE, gaining a precision of 0.993 and an accuracy of 96%, in comparision to the existing results in the literature. With precision and accuracy, we gained a recall of 0.673 and an F1-Score of 0.771, which is better than the average of the other works. The used dataset was attained from ITrust Center, Singapore University of Digital Science, reliable for anomaly detection in an ICS environment.

**ARTICLE HISTORY**
Received 23 January 2024
Accepted 19 March 2024

**KEYWORDS**
Anomaly detection; ICS; security; SWaT; deep learning

## 1. Introduction

Industrial control systems (ICSs) are important in critical infrastructure like energy, dams, communication, water, chemicals, etc. These systems are responsible for process monitoring and control. Monitoring gathers information from multiple sensors and confirms that the processes are operating as intended. In the meantime, the controlling component uses actuators and decision-making algorithms to oversee the processes and start actions. Due to the widespread use of the Internet of Things (IoT) and networked sensor technologies in ICS, critical infrastructures are now vulnerable to various cyber threats and adversary actions. Successful cyberattacks against these vital systems are becoming more common (Aslam et al., 2023; Dehlaghi-Ghadim et al., 2023). Furthermore, abnormal operations caused by hardware malfunctions in critical infrastructure must be promptly detected and intercepted to avoid possible harm to human life or infrastructure damage. For instance, authors presented a typical configuration of an ICS network with a distributed system shown in their Figure 4 (Aslam et al., 2023). The ICS infrastructure is dispersed in this configuration with controllers, sensors, and actuators. A centralized supervisory control is implemented to monitor and manage several remote devices connected to the ICS. The local area or control network dedicated to the ICS associates the centralized control station and the various ICS components. Such industrial control systems are vulnerable to targeted attacks, including cyberattacks, due to their complicated nature. In light of these vulnerabilities, it is critical to investigate techniques for efficiently identifying process abnormalities in ICS. Ensuring the integrity and security of vital infrastructures requires immediate detection and elimination of these anomalies.

Numerous works have been done on anomaly detection in ICS. For time series prediction and analysis, permitting anomalies to be noticed based on values that deviate from goals, the authors used the Long Short-Term Memory (LSTM) model in Khan et al. (2022). To detect anomalies, the authors in Kravchik and Shabtai (2021) employed

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an encoder-decoder model, which entails encoding input data into a latent representation and decoding it back into its original format. Finding anomalies can be done by comparing the reconstruction error between the original and reconstructed data. In Pavel et al. (2020), the authors used Principal component analysis (PCA), dynamic PCA, Fisher discriminant analysis (FDA), dynamic FDA, canonical variate analysis (CVA), and partial least squares (PLS) to identify data points that deviated significantly from the extracted features. Similarly, authors used other techniques like process invariants in Feng et al. (2019) to identify invariant patterns within a process. One-class SVM and segmentation are in Chang et al. (2021), and the clustering technique is in Tomlin Jr et al. (2016). These techniques usually use databases to store the signatures of popular cyberattacks, which they then compare against the monitored traffic. A cyberattack must be seen before, and its signature must be recorded in the database to be recognized. The failure of this approach to identify specialized or previously unseen cyberattacks presents a serious limitation. The approach for anomaly detection has changed to focus on deep learning (DL) and machine learning (ML) techniques to get around such limitations. These methods can potentially provide high-performance detection with shorter evaluation times (Khan et al., 2022). Since cyberattacks are rare and considered anomalies, anomaly detection is frequently used to detect them. Even though several ML/DL-based anomaly detection techniques have been put forth in various fields, successful anomaly detection requires particular considerations due to the unique features of industrial scenarios.

The researchers in Goh, Adepu, Tan, et al. (2017) used a LSTM approach for anomaly detection in ICS using cumulative sum (CUSUM), but focussed on one process P1 (SWaT dataset)of the ICS environment. It is required for anomaly detection in the overall ICS environment. In another work (Inoue et al., 2017), the authors presented a deep neural network (DNN) with a layer of LSTM approach for time dependencies of sensor information. The proposed LSTM-based approach is insufficient for anomaly detection in ICS because of its difficulty detecting gradual changes and anomalous actuator behaviour. Convotuional Neural network (CNN) is mostly useful for image-based datasets like recent researchers in Mendoza-Bernal et al. (2024) worked on image-based anomaly detection using the CNN approach. Similarly, the authors in Boulent et al. (2019) used CNN for anomaly detection in crops using image-based datasets. In another work (Balla et al., 2024), the authors used DL with an enhanced CNN-LSTM-based approach for intrusion detection in SCADA system and obtained results with 95.21% with a recall of 82.898%. This approach suffers from a high positive rate and computational complexity. Similarly, the authors in Inoue et al. (2017) used an unsupervised ML approach for anomaly detection in ICS and obtained a precision of 0.92500 with a recall of 0.699 and F Score of 0.796, this work suffered from accuracy and high false positive rate. In another work (Shalyga et al., 2018), the researchers used GA to achieve a suitable neural network from the SWaT dataset, in which authors obtained an accuracy of 95% with 0.976, F1 0,812, and recall of 0.702. In this work, authors could detect only one anomaly out of eleven. In this work, the detection rate is only 11%, which is very low for a secure ICS environment.

For anomaly detection in ICS → overcome the research challenges like computational complexity, high false positive rate, and accuracy, this work proposes an ImpAE technique. Using the SWaT dataset, the proposed study applied the ImpAE deep learning model for anomaly detection in ICS. This model received three higher-correlated features, as shown in Figure 1 from the dataset and incorporated the experiment.AE does not require any labelled data in contrast to CNN, which requires labelled data (which can be a limitation in an ICS environment). AEs are designed for data reconstruction, and this reconstruction permits the AEs to detect anomalies based on deviations from normal patterns, even without explicit anomaly labels. AE captures both local and global patterns in the available dataset and also provides a direct interpretation of the reconstruction error.

Previously, a review paper by (Aslam et al., 2023) that discussed the issues and challenges in the ICS environment and highlighted the anomalous behaviour in an ICS environment. In this work, the Secure Water Testbed (SWaT) dataset is used due to its wider application. This SWaT dataset comprises about 51 tags, 26 actuators, and 25 sensors, as explained in Table 1. The collection duration was about seven days under normal operation and six days, for which 41 attacks impacted the physical system. Moreover, the regularization term allows the proposed model to be tuned to discover distinct attack subsets at the expense of misclassifying instances of standard data or vice versa. Precision, recall, and F1-score were used to evaluate the model's performance, and it was compared with other state-of-the-art anomaly detection techniques. The major contributions of the proposed work are reported below:

- A new ImpAE-based model for anomaly detection yields from cyberattacks affecting ICS security. Using deep learning techniques, ImpAE-based anomaly detectors can be integrated with another detector specific to another anomaly.
- Develop an approach for fine-tuning the model based on particular industrial requirements and relevant

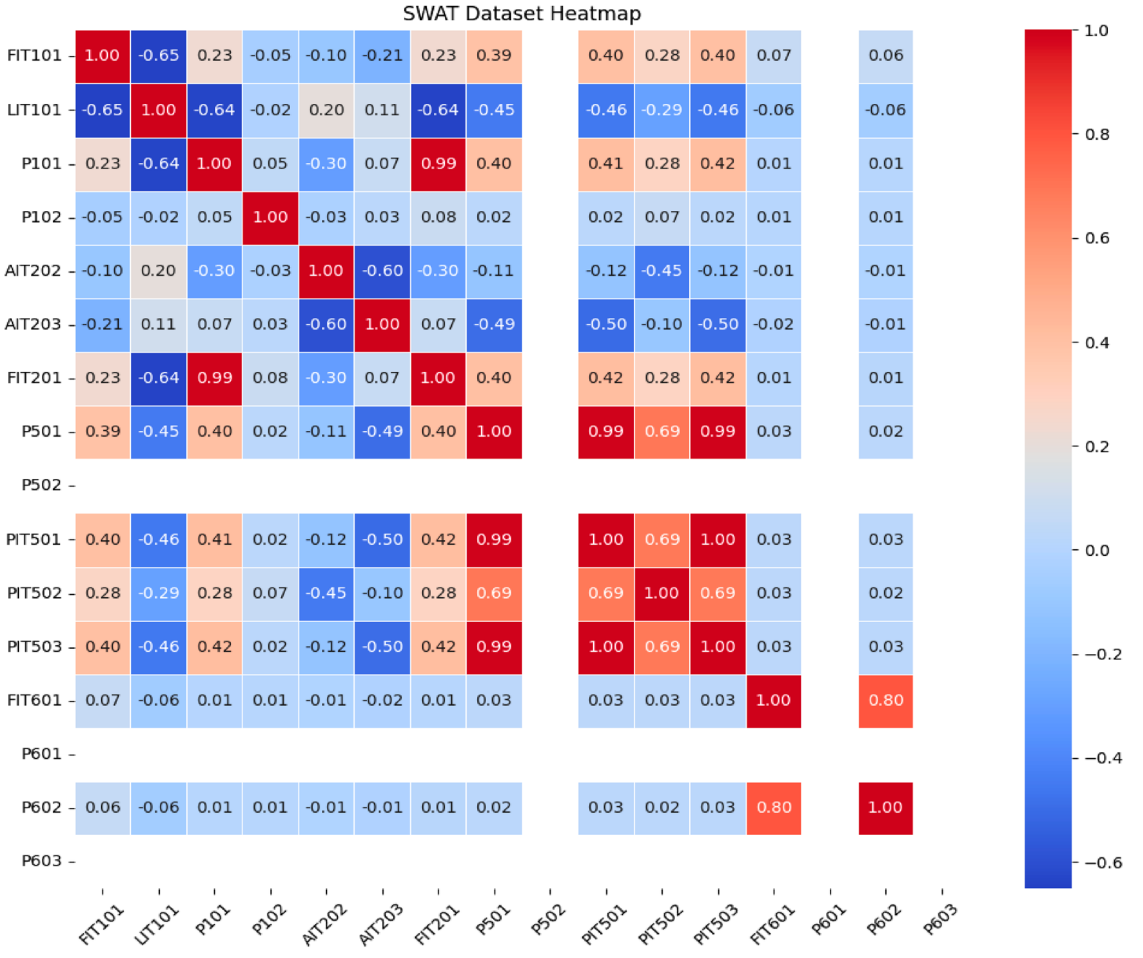



**Figure 1.** Statics of the data set: Highly correlated features $= 1$; Highly inverse correlation $= -1$.

performance parameters by incorporating the other regularization term. The article proposed a practical method for identifying irregularities in an ICS environment.

- A collection of security-focussed functionalities that capture resource utilization patterns to model the behaviour of industrial sensors and actuators regarding security. Cyberattacks that affect resource usage may affect those characteristics and, in turn, industrial process security.
- Validating the proposed approach through a set of experiments using a real-world ICS dataset (SWaT) show that the presented approach performs better than some state-of-the-art models and has a swift training process.
- Cleaning dataset with ML techniques. Among the tasks considered in this step are feature scaling, corrupt and missing value treatment, dataset splitting, and categorical feature encoding. Selects only the most discriminating features through filtering. Features that show abnormal behaviour, high correlation, or zero variance are eliminated.

The rest of the paper, we organized as; Section 2 is about Research background and motivation, introducing ICS and deep learning techniques used for anomaly detection in ICS. Section 3 is about Material and Methodology in which we explore the used data set, SWaT environment and communication, attack scenarios, and methodology used in the proposed work. Section 4 is ICS anomaly detection and experimental configuration. Section 5 is results and discussion, Section 6 is discussion and future work, while in Section 7, we concluded our work.

## 2. Research background and motivation

This section demonstrates the introduction of ICS and cyber attacks, defense models for ICS, different deep learning models used for attacks and defense in ICS, metrics, and available and used datasets. We studied the deep learning-based anomaly detection models in ICS, and lastly, we briefly provided an overview of the motivation for this work.

**Table 1.** List of serving sources (Sensors and actuators) in SWaT dataset according to Goh, Adepu, Junejo, et al. (2017).

| Sr.No | Traits | Influence | Category |
|---|---|---|---|
| 1 | FIT-101 | Flow meter | S |
| 2 | LIT-101 | Transmitter | S |
| 3 | MV-101 | Overflow (tank)- Motor valve | A |
| 4 | P-101 | To send water from P1 to P2 | A |
| 5 | P-102 (Feedback) | It's a pump for feedback process from P1 to P2 | A |
| 6 | AIT-201 | NaCl level measuring | S |
| 7 | AIT-202 | HCl level measurement | S |
| 8 | AIT-203 | NaOCL level measurement | S |
| 9 | FIT-201 | Control pump or flow transmitter | S |
| 10 | MV-201 | Motor valve for controlling UF water feed tank | A |
| 11 | P-201 | NaCl metering pump | A |
| 12 | P-202 (Feedback) | NaCl metering pump | A |
| 13 | P-203 | HCl metering pump | A |
| 14 | P-204 (Feedback) | HCl metering pump | A |
| 15 | P-205 | NaOCl metering pump | A |
| 16 | P-206 (Feedback) | NaOCl metering pump | A |
| 17 | DPIT-301 | Control the backwash process using a differential pressure indication transmitter. | S |
| 18 | FIT-301 | The flow meter gauges the water flow within the UF stage. | S |
| 19 | LIT-301 | The level transmitter monitors the water level in the UF feed water tank. | S |
| 20 | MV-301 | The motorized valve regulates the UF backwash process. | A |
| 21 | MV-302 | Motorized valve: Controls UF-to-DeChlorination flow. | A |
| 22 | MV-303 | Motorized valve: Manages UF backwash drainage. | A |
| 23 | MV-304 | Motorized valve: Manages UF drainage. | A |
| 24 | P-301 (Feedback) | UF feed pump: Moves water from UF tank to RO tank via UF filtration. | A |
| 25 | P-302 | UF feed pump: Moves water from UF tank to RO tank via UF filtration. | A |
| 26 | AIT-401 | Water RO hardness meter | S |
| 27 | AIT-402 | ORP meter: Regulates NaHSO3 dosing (P203) and NaOCl dosing (P205). | S |
| 28 | FIT-401 | Flow transmitter: Manages the UV dechlorinator. | S |
| 29 | LIT-401 | The level transmitter monitors the water level in the RO feed water tank. | A |
| 30 | P-401 (Feedback) | Pump: Transfers water from RO feed tank to UV dechlorinate. | A |
| 31 | P-402 | Pump: Moves water from RO feed tank to UV dechlorinator. | A |
| 32 | P-403 | Sodium bisulfate pump. | A |
| 33 | P-404 (Feedback) | Sodium bisulfate pump. | A |
| 34 | UV-401 | Dechlorinator: Eliminates chlorine from water. | A |
| 35 | AIT-501 | RO pH analyser: Determines HCl concentration. | S |
| 36 | AIT-502 | RO feed ORP analyser: Monitors the level of NaOCl. | S |
| 37 | AIT-503 | RO feed conductivity analyser: Monitors the level of NaCl. | S |
| 38 | AIT-504 | Permeate conductivity analyser: Monitors NaCl concentration in RO permeate. | S |
| 39 | FIT-501 | RO membrane flow meter: Measures flow rate of water entering the RO membrane. | S |
| 40 | FIT-502 | RO permeate flow meter: Measures the flow rate of RO permeate. | S |
| 41 | FIT-503 | RO reject flow meter: Measures the flow rate of RO reject water. | S |
| 42 | FIT-504 | It Measures the flow rate of water in the RO recirculation system. | S |
| 43 | P-501 | Pump: Transfers dechlorinated water to RO. | A |
| 44 | P-502 (Feedback) | Pump: Moves dechlorinated water to RO. | A |
| 45 | PIT-501 | RO feed pressure gauge. | S |
| 46 | PIT-502 | RO permeate pressure gauge. | S |
| 47 | PIT-503 | RO reject pressure gauge. | S |
| 48 | FIT-601 | UF backwash flow meter. | S |
| 49 | P-601 | Pump:Transfers water from RO permeate tank to raw tank (not for data collection). | A |
| 50 | P-602 | Pump: Moves water from UF backwash tank to UF filter for membrane cleaning. | A |
| 51 | P-603 | No implementation in SWaT yet. | A |

Explanation; A = Actuator and S = Sensor

### 2.1. Industrial control system

Field sites are an essential component of ICSs, as they monitor and regulate industrial processes (Aslam et al., 2023). ICSs are composed of control centres. This location includes PLCs (programmable logic controllers), actuators, and sensors. The PLC comprises a power source, input and output modules, memory for storing control logic programs, and a basic structure. Sensor values are sent to the input module, which uses them to compile data regarding the ICS's current state. The PLC then uses control logic routines to communicate with the actuator and manage the physical process. Due to various security threats, several PLCs and hardware components in ICSs nowadays lack self-protection capabilities. Deploying IDSs in ICSs is, therefore, imperative. However, their straightforward form makes protecting ICSs at the PLC level difficult. Moreover, each PLC can only monitor the hardware it controls, which limits the detection efficiency of the IDS installed on the PLC. IDSs are, therefore, typically installed on bypasses of industrial control networks.

The most common ways to carry out attacks on ICSs are via inserting malicious code and manipulating sensor or actuator settings. The incongruity between the control signals, or actuator values, and the system states, or sensor values, indicates an attack because of the aberrant control logic. For instance, the reaction pipeline will break if the gas injection operation occurs while the gas pressure is almost at the top threshold. In an attack of this kind, the mismatch between the injection operation and the gas pressure reflects the anomalous control logic. However, as industrial control logic gets more intricate, it gets harder for specialists to create efficient rules for attack detection.

In the days of conventional ICS, safety was the top priority in a safe, closed environment. Every stage of the system was built with the ability to stop operations if anomalies were found. On the other hand, introducing Internet protocols, wireless technologies, and Internet of Things (IoT) devices into ICS has resulted in less isolation from outside influences. Regrettably, this led to multiple vulnerabilities because the ICS environment's security measures were insufficient.

Field sites and control centres are usually vulnerable to intrusion in an ICS environment. Industrial processes are controlled and observed at field sites using sensors, actuators, and Programmable Logic Controllers (PLCs) (Alanazi et al., 2023). A PLC comprises three basic components: input and output modules, power supply, and memory that stores the control logic program. In an ICS, the input module system gathers sensor data or values to determine the process's current state. The control logic program regulates the physical processes by sending signals to the actuators. Consequently, we present a novel IDS in this work based on the rules we created by analysing the information that the deep learning-based models had acquired.

Figure 4 in Aslam et al. (2023) is for the components and structure of ICS and various levels (Physical process L0 to L5) in the ICS environment. The working explanation of the various levels is given as

L0 → At this level, actuators, sensors, and instrumentation components directly connect to and control the manufacturing process. L1 devices manage these components.

L1 → This level includes RTUs, PLCs, and IEDs, among other intelligent devices. It is in charge of gathering data from L0 and comprises process control mechanisms.

L2 → Control systems like HMI, control rooms, and alarms are present at this level.

L3 → Manufacturing operation systems are included in this level, which also oversees plant control for production. It includes network file servers, data historians, engineering workstations, remote access servers, and IT services like DNS and Active Directory.

L3.5 → This level includes firewalls and proxies, among other security systems. It entails keeping the IT and ICS environments apart. At this level, the components are remote access servers, patch management, and intrusion detection systems.

L4 → This level consists of an IT system that deals with printing, operational, email, and phone services. L5 → Applications and IT infrastructure systems are at this level. L2 and L3 can communicate by accessing the enterprise zone via the DMZ.

- In addition to maintaining safe connections between OT and IT networks, the DMZ enables data exchange between the enterprise and control zones. Using several DMZ firewalls, it uses a secure network architecture.
- Traditional IT devices and systems, including enterprise networks and logistic business systems, are included in the IT network, also known as the enterprise zone.

ICS networks were not previously subject to security-related monitoring. Rather, ICS systems were usually protected from outside attacks by installing a firewall between L4 and L5. The physical processes could not be manipulated by compromised devices at higher levels thanks to this isolation. Nonetheless, the risk of compromise has increased dramatically due to ICS environments being more exposed to the Internet and outside

parties. An attacker can alter data transmitted over the network to cause process failure or degradation if they can access portions of an ICS located in L0 through L2. The BlackEnergy (2015) and Industroyer (2016) attacks on the Ukrainian power grid were notable examples of this strategy in action [20], which caused a power outage that affected over 200,000 people for several hours. A chemical processing plant was forced to close due to the Triton malware attack in 2017 [6]. Thus, it is essential to watch for any signs of possible compromise or misuse of ICS networks through active monitoring.

The primary focus of this work is on techniques that involve training machine learning (ML) models with deep learning capabilities to perform reconstructions of systems and detect possibly anomalous states of systems based on high reconstruction errors [18, 32, 37]. We focus on deep-learning model because it is better to detect anomalies than conventional methods [25].

### 2.2. Deep learning-based defense models in ICS

This section highlights the DL-based defense models generally used in ICS environments. Intrusion detection system (IDS) models in ICS use machine learning algorithms to identify and categorize cyber threats by monitoring network traffic, system logs, and other data sources. By learning typical behaviour patterns, these models can recognize abnormal activity that might point to an ongoing cyberattack. LSTM networks are useful for analysing sequential data in ICS security, like network packet sequences or system logs. With time, they can pick up patterns and behavioural traits that allow them to be used for anomaly detection, malicious activity identification, and historical data-driven system failure prediction. Unlike AEs, unsupervised learning models try to reconstruct the given data. The components of these systems are an encoder and a decoder. The encoder converts the input data into a latent space, a lower-dimensional representation, while the decoder uses the latent space to reconstruct the original input. AEs can help the model reconstruct typical behaviour and identify deviations as anomalies when used for tasks like anomaly detection in ICS security. The AE can identify new or unknown threats based on reconstruction errors by training them on normal data. CNNs are often utilized for tasks related to pattern recognition and image analysis. CNNs can be used to analyse log files or network traffic data in the context of ICS security, where the data can be represented as multi-dimensional arrays. CNNs are useful for tasks like malware classification, image-based anomaly detection in ICS environments, and intrusion detection because they can automatically extract pertinent features from the data.

The authors in Liu et al. (2024) proposed a defense and attack approach based on LSTM-ED (encoder-decoder). The authors equalize reconstruction errors across various features to ensure strong defense outcomes and prevent anomaly scores from being impacted by poorly predicted features. They used three datasets from SWaT, WADI, and BATADAL for experimental setups. The outcomes obtained a precision of 0.6626. The authors in Ni and Li (2024) highlighted the machine learning-based security issues in the IIoT environment. Similarly, in another work (Bahadoripour, 2024), the researchers used Explainable DeepFederated Multi-Modal for attack detection in ICS environment. In this work, the authors enhanced the accuracy of attack detection in ICS and the interoperability of cyber-attack detection.

### 2.3. Deep learning based anomaly detection techniques in ICS environment

This section is about DL-based anomaly detection in ICS. Generative Adversarial Networks (GANs); Generator networks and discriminator networks, which play a min-max game, make up GANs (Adiban et al., 2023). To train a generator to produce samples that closely resemble the normal data distribution, GANs can be used in anomaly detection on regular ICS data (Adiban et al., 2023). After that, the discriminator is employed to distinguish between the produced samples and the actual data. Anomalies are identified when the discriminator cannot distinguish generated and real samples. AEs are generative models capable of autonomously learning the normal data's underlying distribution (Tabassum et al., 2024). They comprise a decoder network that reconstructs the input from the latent space and an encoder network that maps the input data into a latent space. AEs can learn to recognize normal patterns in ICS data and recognize deviations as anomalies. Because Recurrent Neural Networks (RNNs), such as LSTM networks, are good at modelling sequential data, they can be applied to identifying anomalies in time-series data, like system logs or readings from ICS sensors. An RNN can learn temporal dependencies and spot abnormal behaviour by training on historical data. CNNs are useful for jobs like malware classification, anomaly detection in ICS environments, and intrusion detection because they can automatically extract pertinent features from the data.

### 2.4. Related work in ICS anomaly detection

This section highlights the literature review in ICS anomaly detection using various generally used detests. Authors in Myers et al. (2018) presented a process mining approach to anomaly detection for attack detection and

anomaly detection identification in an ICS environment. The authors used a conformance-checking analysis technique and ICS data logs. The authors presented experimental logging for ICS devices. However, the provided research is very limited to anomaly detection in ICS using process mining. The authors in Gómez et al. (2019) provide the Electra Dataset, suggest a process for creating trustworthy anomaly detection datasets for ICS, and show how well Machine Learning and Deep Learning models can detect abnormalities using the dataset. In another study (Mokhtari et al., 2021), authors presented a unique method utilizing supervised machine learning models to identify intrusions using SCADA system data: the measurement intrusion detection system (MIDS). The outcomes show how successful the suggested method is; random forest outperforms other classifiers in identifying abnormalities in the testbed, but effective intrusion detection systems are required in ICS environments to identify and categorize anomalous actions, especially covert attacks, using measurement data from the supervisory control and data acquisition (SCADA) system. The authors in Tekeoglu et al. (2021) offered an innovative approach to real-time cyber-physical network traffic-based intrusion/anomaly detection for Industrial Control Systems (ICS). A specific evaluation of the approach was carried out utilizing a network trace from a (SWAT) at the iTrust Laboratory in Singapore. In Hwang and Lee (2021), authors used the Electra dataset for anomaly detection in ICS. The authors used various ML and DL models to identify anomalies in the ICS environment. In Abdelaty et al. (2021), authors introduced DAICS, a cutting-edge deep learning framework created for big ICSs that aims to lower false alarms in anomaly detection while addressing the difficulty of changing normal behaviour. The system integrates an automatic threshold tuning method for detection accuracy and uses a 2-branch neural network to learn ICS behaviour changes with few data samples. Still, several challenges exist to overcome anomaly detection in an ICS environment. Authors in Wang et al. (2021) studied a deep residual CNN-based model for anomaly detection in ICSs. The model tackles the lengthy training process of deep learning algorithms and the shortcomings of conventional machine learning techniques in identifying known assaults. It ensures the safety of ICSs by using transfer learning and image conversion algorithms to achieve accurate predictions for unknown or differentially distributed aberrant data. In Fung et al. (2022), authors thoroughly assess prior reconstruction-based anomaly detection methods for ICS. The research also emphasizes how commonly used criteria for ML algorithm evaluation are inadequate for anomaly identification in ICS setups. The work aims to better capture ICS-specific objectives by introducing range-based metrics, which will help ICS security practitioners choose and configure reconstruction-based anomaly detectors more effectively. In Boateng et al. (2022), authors used the SWaT dataset for anomaly detection in ICS and used a class neural network model. The authors in Adepu and Mathur (2018) suggested a distributed assault detection approach to detect anomalous factory operations or physical behaviours. This approach resulted in the creation of a set of guidelines based on the invariants of the factory's manufacturing procedure. However, creating these rules by hand made identifying attacks involving intricate control logic or industrial operations difficult. Authors Adepu and Mathur (2018) recognized the challenge of figuring out the industrial chemical invariants–like the reagent concentration–and developing rules based on those reactions. As machine learning has advanced, more researchers are concentrating on combining ML-based techniques with commercial intrusion detection systems (Elmrabit et al., 2020; Liang et al., 2019). Presented a one-class graphical classifier that depends on ICSs' typical behaviour (Elmrabit et al., 2020). This classifier makes identifying anomalous sensors and actuators possible using the interpretability of graphical models. However, fulfilling the condition of variable independence in Naive Bayes classifiers is difficult because of the interconnectivity between sensors and actuators in ICSs. Based on a one-class support vector machine (OCSVM), the authors in Inoue et al. (2017) created an intrusion detection model that considers the acceptable physical states of ICSs.

Deep learning has become more common in intrusion detection techniques in recent years. To detect attacks on ICSs, in Shalyga et al. (2018) and Aldweesh et al. (2020), authors created a DNN and one class SVM-based model for anomaly detection using the SWaT dataset. This work is limited to deliberately injected anomalies instead of real-life anomalies and to the SWaT dataset. Moreover, a higher recall rate is required to enhance the overall performance of ICS security. An intrusion detection technique that calculates the discrepancy between observed and expected values of industrial elements was presented by authors in Abdelaty et al. (2021) and Tama et al. (2022). Neural networks of many kinds, including recurrent and convolutional neural networks (CNNs and RNNs), can be used with this technique.

The intrusion detection framework presented by the authors in Quincozes et al. (2021), Pashaei et al. (2022), and Süzen (2021) uses greedy algorithms and deep and broad neural networks (DNNs). With the help of this enhanced framework, settings can be updated automatically and without human involvement.

Based on the literature review, previous researchers highlighted various ML and DL techniques for anomaly

detection in ICS, but none of these techniques cover the overall model architecture. There is only one research work (Fung et al., 2022) that tried to cover the partial selection of model architectures. In this work, the authors still didn't resolve the accuracy and computational complexity issues, making it unclear how to implement the applied approaches. Existing works require higher computational power. Until now, there is no work to show the generic technique for anomaly detection in ICS. Existing techniques have no standard criteria, which results in complications. Without a standard methodology, the researchers' findings could be deemed illegitimate if they followed the incorrect procedures. In our recently published review (Aslam et al., 2023), we found that in existing literature, the techniques used require more data for training to yield suitable results. To overcome these research gaps, the proposed model yields suitable and friendly results for the research and practical environment.

## 3. Materials and methodology

In this section, we will explain the dataset and methodology utilized. This particular section has been divided into seven distinct subsections.

### 3.1. Secure water testbed (SWaT) dataset

According to iTrust (2016), SWaT was launched on March 18, 2015. The basic purpose of this lab was research in cyber security, and it was established with the coordination of the Singapore Ministry of Defense. The project's objective was to secure the ICS environment from cyberattacks. For the virtual visit of SWaT, one can visit the iTrust Centre In Cyber Security Research (2016).

#### 3.1.1. Secure water testbed (SWaT) architecture

As seen in Figure 2, SWaT is a fully operational scaled-down water treatment plant due to its miniature size and capacity to produce five gallons of double-filtered water per minute. This testbed is a model of large-scale, contemporary water treatment facilities that usually exist in cities. Its main goal is to support experimentally validated research to develop safe and secure ICS architectures. Six essential processes are included in SWaT, which are related to the facility's control and physical components.

P1 → Raw water and tank storage

P2 → This step assesses water equality, also known as the pre-treatment process. Chemical dosing is carried out when the water quality exceeds acceptable limits.

P3 → At this stage, filtration membranes are used to remove undesirable materials.

P4 → Once the residuals have undergone filtration via the Ultra Filtration System (UFS), residual chlorine is eliminated through the dechlorination process using ultraviolet (UV) lamps.

P5 → Water from P4 is pumped to stage P5, known as the reverse Osmosis process.

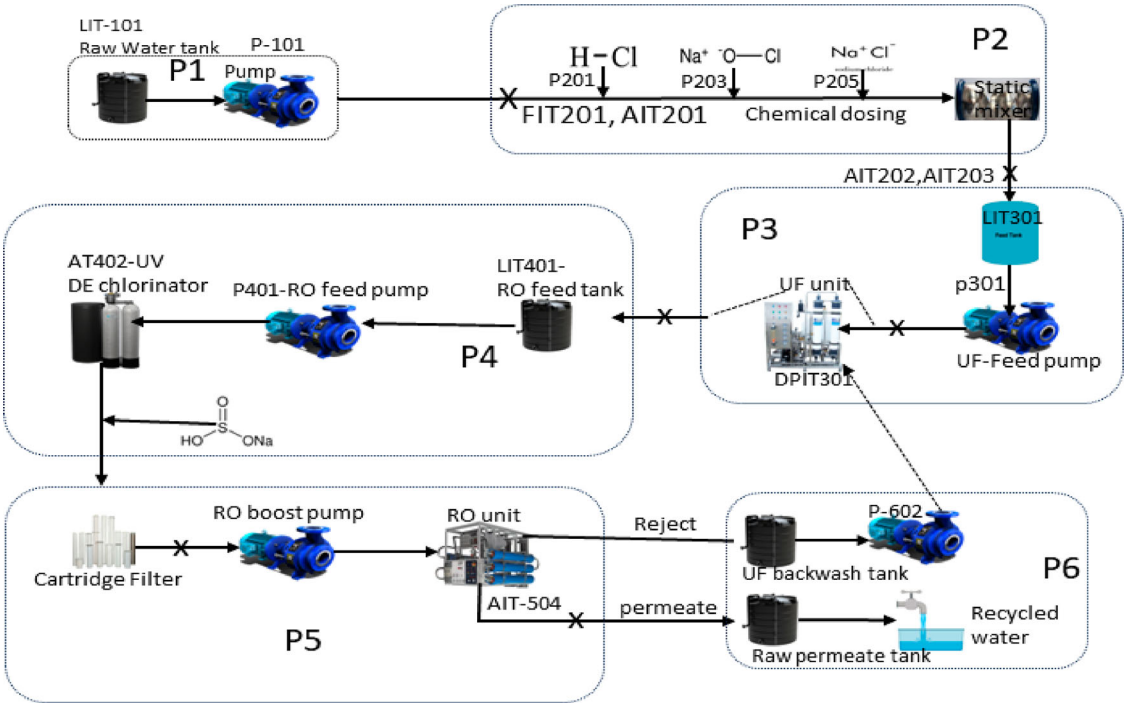



**Figure 2.** Process view of SWaT testbed.

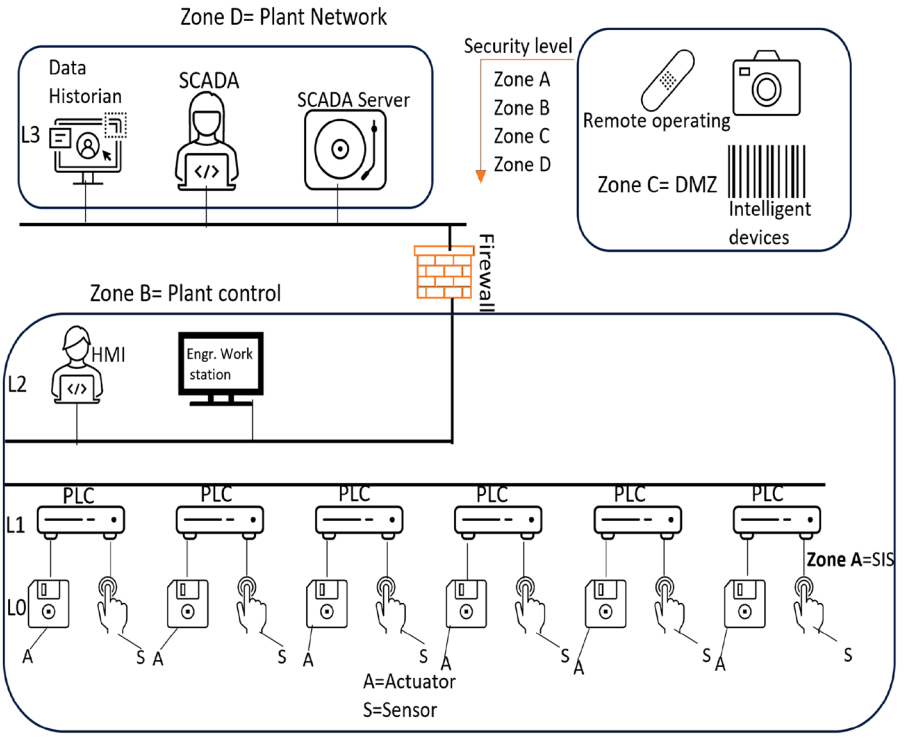



**Figure 3.** Architicture of SWaT Testbed.

P6 → In the concluding stage, the remaining water is stored in a separate tank, which can subsequently be directed to step three for the backwash process.

### 3.1.2. SWaT communication

For communication purposes in the SWaT environment, it consists of various zones of the communication network, as explained in iTrust (2016) and shown in Figure 3, like Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA) Historians, and Human Machine Interfaces (HMI). The SCADA system has access to data collected from the sensors, which is then recorded by the historian for further analysis. SWaT consists of two levels, which are given as L0 → This level comprises a ring network that facilitates the transmission of actuator and sensor data to the respective PLCs.

L1 → This level is like a star network configuration, enabling communication between the SCADA system and the six dedicated PLCs, each responsible for a specific process.

The communication between actuators, sensors, and PLCs is via wireless or wired links. It depends on the manual switch mode or permission (Figure 4).

## 3.2. Attack scenarios

There were 41 cyberattacks used in the SWaT testbed. Most of these compromises aimed to breach the system's cybersecurity, attempt to stop the service or provide water at a slower rate than necessary. But out of all the attacks, we found nine that compromised the system (numbers 6, 11, 19, 20, 22, 24, 28, 38, and 40 in [31]). The breaches specifically targeted UV lights, chemical dosing sensors, or actuators. Below is a list of these cyberattacks.

### 3.2.1. Attack on water quality

AIT202 is a sensor for measuring the HCl level in the second process (P2). It is affected by attack 6. The approximate time for this attack is about 4 minutes. The objective of this attack is to reduce water quality.

### 3.2.2. Energy desruption

FIT401 is a sensor with fixed values for 0–9 minutes. The attack target is energy disruption by the shut-down process of the UV actuator. It disturbed water distribution. It is noted at attack 11.

### 3.2.3. RO process disruption

AIT504 is a chemical sensor that monitors the NACl in the sixth-process (P6) tank. Attacks 19 and 20 adjusted this sensor to an unchanged value of 16 for a few minutes. The objective of the attack was to disrupt the RO process. In attack 20, the value changed from 16 to 255 minutes and time from 4 to 6 minutes, respectively.

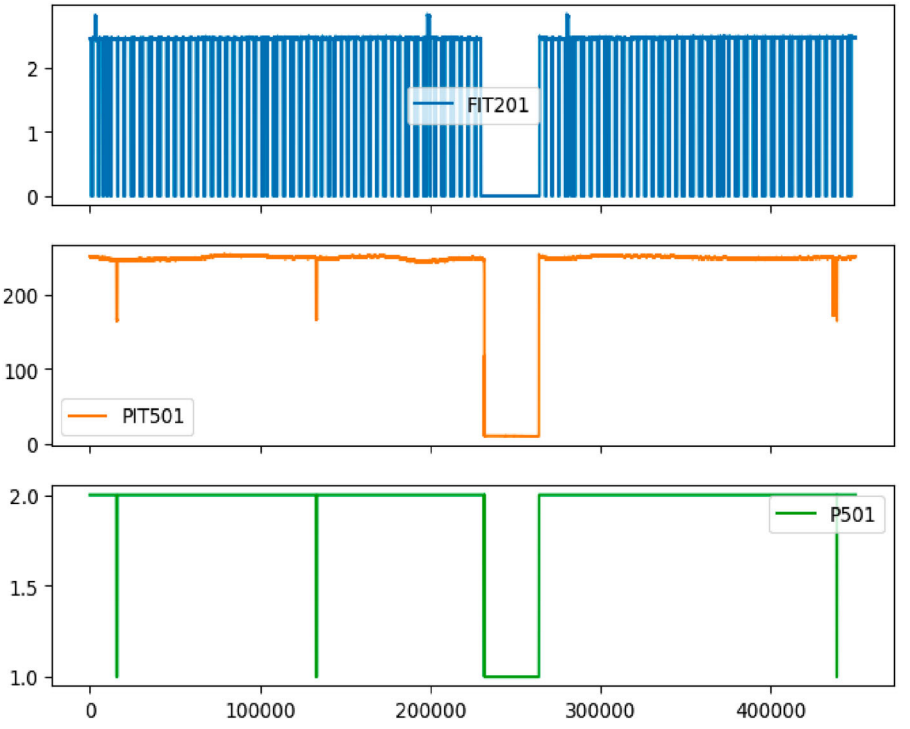



**Figure 4.** Observations from FIT201, P501 and PIT501 (Initial stage observations).

#### 3.2.4. RO process damage

Here, attack 22 disturbs the various sensors and actuators, specifically UV401, known as the lamp actuator, AIT502, and P501. AIT502 monitors the NaCl level while the pump sends dechlorinated water to reverse osmosis P501. Attack 22 was for about eight minutes, blocked the UV lamp, and fixed a fixed value of 150 at AIT502. Also, it was forcing P501 to remain unchanged. Due to their similar attack point and near-constant occurrence, these were indistinguishable from the developed model. The overall attack objective was to affect the RO process badly.

#### 3.2.5. Disturbing water quality

Attack 24 was to target P203, which is in charge of HCl dosing, and P205, which is in charge of NaOCL dosing. Both HCl and NaOCL greatly affect water quality, so this attack aimed to disturb the water quality.

#### 3.2.6. Water drainage

The attack 38 affected the AIT402 in fourth process (P4) and AIT502 in fifth process (P5) to vary the HCl and NaOCL respectively. The attack lasted about five minutes, and both sensors were changed to 260 to drain the water. The basic objective of the attack was to drain the quality of the water.

#### 3.2.7. Shutdown UV process

Attack 40 was to affect the sensor FIT401 in P4. The cyberattack, which lasted for five minutes, involved adjusting the FIT401 to 0. Like cyberattack 39, its objective is to stop UV and direct water to RO processes.

The data was continuously logged onto a Historian server every second. Historian server data was gained from the testbed actuators and sensors. Sensors are used to convert physical properties into electronic output, while actuators are used to convert signals into physical properties like turning the pump on or off. This dataset is for the operational mode physical properties. Within 51 attributes, as discussed in Table 1, and six processes, as discussed in Figure 2, 947,722 samples were collected in eleven days. Cyberattacks can be profiled using data that captures their physical characteristics.

### 3.3. Data pre-processing

SWaT dataset consists of two .csv files. Out of these two files, one contains the normal behaviour values of sensors and actuators and data collected for a few days. Another file consists of cyber attacks which has been noticed for four days and consist of normal and attacked behaviour. We extracted our training and validation datasets using the one-week normal behaviour file. Specifically, we divided the available dataset with normal behaviour into the first 80% of the samples for training, 20% for the validation dataset, and the remainder of the dataset. Conversely, our test dataset consisted of a four-day file with typical and malicious behaviour.

Investigating the dataset is essential to identify false or faulty values for typical behaviour. We plotted the data in this step using programs like Matplotlib and Seaborn. After analysing the values of each characteristic, we concluded that the first records in the collection matched the ICS warm-up phase. Depending on the feature being studied, the warm-up process takes a different amount of time. Figure 7 is after the warm-up process of the FIT201 parameter, and Figure 10 represents the stabilizing time for three parameters: FIT201, P501, and PIT501. Previously, authors used only two features and removed some important values from two features (Perales Gómez et al., 2020). After cleaning the data from attacks, the remaining data was considered normal. To become normal, the ICS environment takes some time.

### 3.4. Filtering

After finishing the first stage of feature filtering, we performed a covariance analysis to ensure there was little correlation between the features and the label. Then, we ran a variance analysis for every feature in the dataset to find the ones that didn't change. Furthermore, we investigated whether the features from two datasets–the distribution was identical for both samples–the first with only normal behaviour and the second with both normal and malicious behaviours, in keeping with standard procedures in ICS scenarios. An example with three features plotted, FIT201, P501, and PIT501, is shown in Figure 5. Values from the test dataset and training dataset have been shown in Figure 5. There were minor differences between the values in the training dataset and the test dataset's values representing typical behaviour.

A high correlation was shown by some feature pairs in the covariance matrix shown in Figure 1. In this statistic, the direction and strength of the correlation are indicated by the cell's colour. Cooler colours, like blue, indicate negative correlations, while warmer colours, like red, indicate positive correlations. Darker colours indicate stronger correlations, and the colour's intensity indicates the correlation's magnitude. Notably, with a correlation of 99.99%, P501 and PIT501 were the features with the strongest correlation. This correlation makes sense, given that pump P501 transfers PIT501 (dechlorinated water) to P5. These correlations are typical in ICSs where specific sensors or actuators are interdependent. With a correlation of roughly 80.33%, PIT503 was the feature with the highest correlation between the label and each feature. On the other hand, less than 10% of the chemical sensors, including AIT203, AIT202, LIT101, FIT101, and P102, showed correlations with the label. We decided to leave all features in place due to the moderate correlation between the features and the labels, as shown in Figure 6. For the second task, a variance study was carried out, and the results showed that features with smaller variances were usually associated with features having a smaller range of possible values. These features mostly consist of categorical characteristics associated with motorized pumps and valves. We then eliminated those features. On to the third task: we used the K-S test to see if continuous features from the training and test datasets originated from the same statistical distribution. The features of the chemical process sensor showed higher K-S statistics. As a result, we removed features whose K-S statistic score was higher than 0.25. Additionally, as shown in Figure 7, we eliminated the FIT201 70% feature because of notable behavioural differences between the training and test datasets.

### 3.5. Extraction

Feature extraction is the process of obtaining higher-order features from the original features by applying expert knowledge. This section describes the tasks necessary to get those higher-order features since ICSs are intended to handle industrial and critical processes, often including repetitive actions. For example, an ICS that monitors and controls a water tank must consider these limitations to adjust the input and output water valves in reaction to lower and higher water tank variations. We advise utilizing autocorrelation and the Fourier Transform (FT) to extract features to identify these recurring actions. The first step is to extract higher-order features using an autocorrelation function. The delay function k(log) is autocorrelation. In this work, we used correlation over a time window of $L = 200$ (It shows the continuous time steps used as input to the neural network). The model captures temporal dependencies and patterns that can be suspicious of anomalies by considering a data window instead of individual data points. Equation one represents the basic concept of autocorrelation. Further hyperparameters used in work have been detailed in Table 2. In this Table 2, the dense layer is used for capturing the patterns and complex relationships in the dataset. The number of layers 2 represents the depth of the proposed neural network model, while AE layers are used for unsupervised anomaly detection. In the proposed work, we used 3 AE layers to represent the depth of the proposed model. The activation function is useful for learning the neural network's nonlinearity and complex relationships. Since sigmoid functions offer a measure of abnormality, they can be useful for anomaly detection by squashing the output values between 0 and 1. The number of neurons in each layer determines the neural network model's capacity and complexity. The model

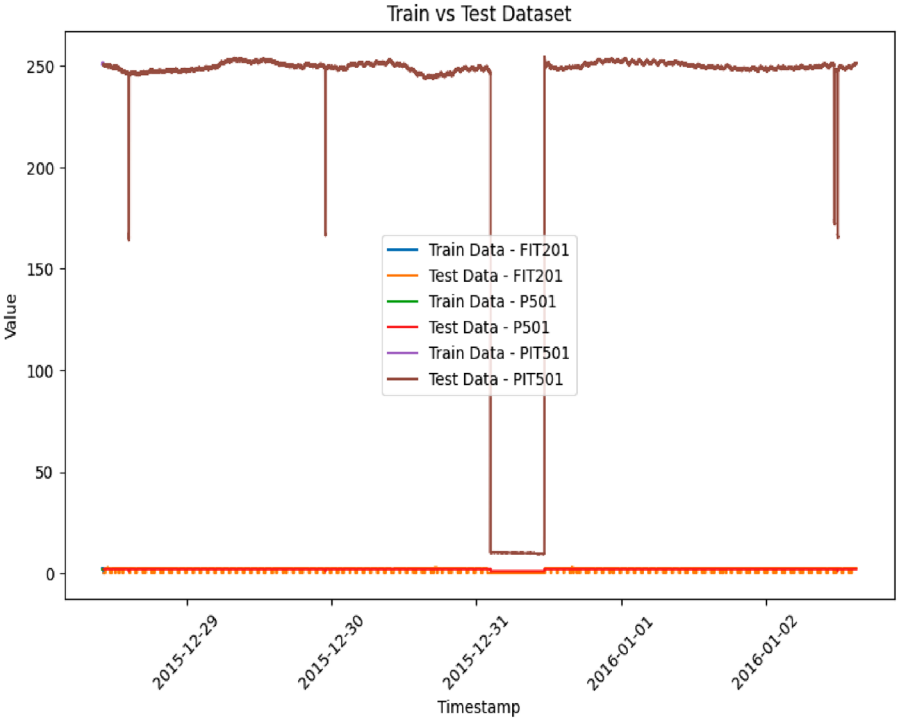



**Figure 5.** Parameter values over training and testing time.

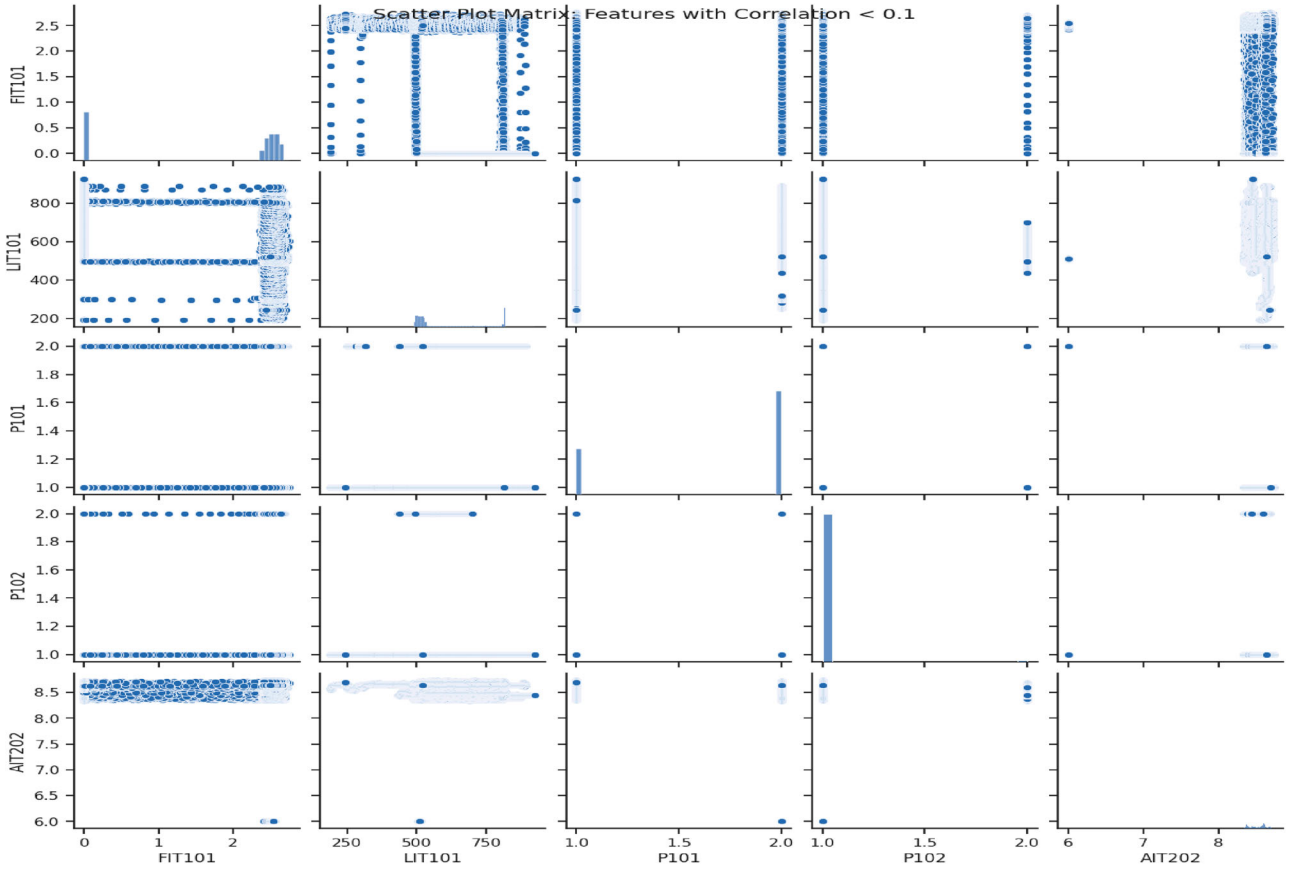



**Figure 6.** Parameters with correlation ¡0.1.

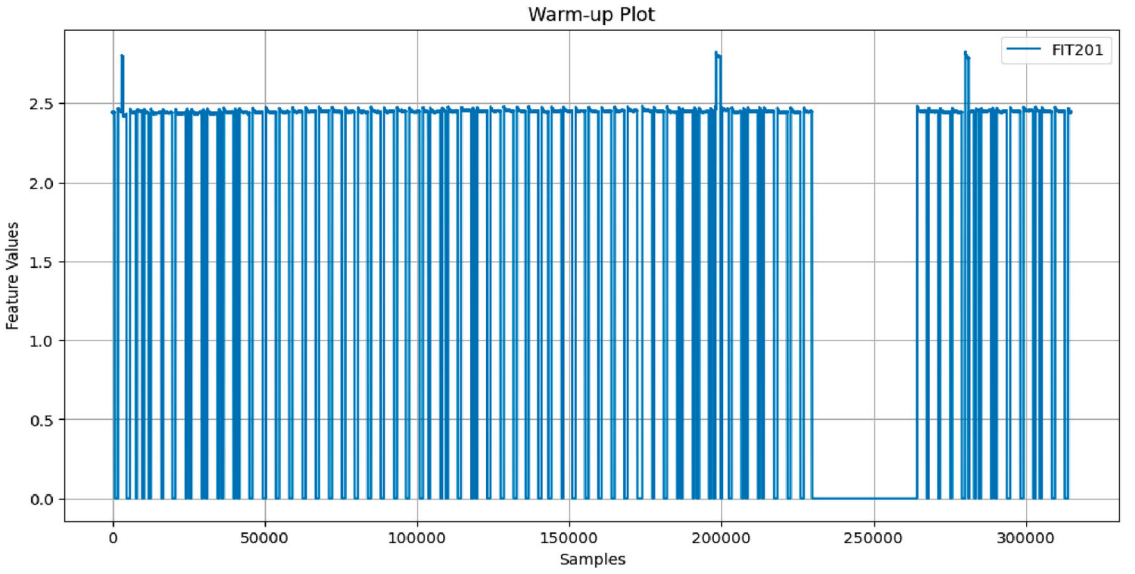



**Figure 7.** FIT201 parameter values over time after removal of 30% samples in warm-up process.

**Table 2.** Hyper parameter used in model (bold).

| Sr. No | Hyper-Parameter | Description |
|---|---|---|
| 1 | Dense layer | 2 |
| 2 | AE layers | 3 |
| 3 | Activation function | Sigmoid |
| 4 | Window length | [150–**200**] |
| 5 | Neurons per dense layer | [**512**–1024] |
| 6 | Neurons per AE layer | [**256**,512] |

can learn more complex data representations with higher numbers of neurons (in this case, 256 for AE layers and 512 for dense layers).

$$\text{autocorr}_{(a_{\rho,G})} = \frac{\sum_{m=\rho-A+1}^{\rho-G}(a_m - \hat{a})(a_{m+G} - \hat{a})}{\sum_{m=\rho-A+1}^{\rho}(a_m - \hat{a})^2} \quad (1)$$

In this Equation (1), $(a_\rho)$is representing the a value at $\rho$ instant while $\hat{a}$) is meant to be findings of repeated patterns, a general example of autocorrelation.

To extract the higher-order feature, we use FT because it is useful in converting time-domain signals into frequency signals. We used the Discrete Fourier transform for this purpose, which is mathematically given as

$$\text{DFT}_{(a_{\rho,G})} = \sum_{n=\rho-A+1}^{\rho} a_{(n^e - 2\pi m/AG(n-(\rho-W+1)))} \quad (2)$$

A signal's periodicity concerning its constituent features can be analysed using DFT and autocorrelation. For every feature, both techniques yield a vector of values. It is advisable to apply statistical functions to each vector, such as the range (the gap between the highest and minimum values), minimum, maximum, standard deviation, and mean to describe the findings. Five features are generated from the DFT analysis and five from the autocorrelation analysis, totaling ten additional features that can be added to the dataset. Figure 8 shows that feature AIT202's variance is almost zero, while feature AIT203's variance is about 50%. this Figure 8 is representation of filtered dataset.

### 3.6. Fine tuning and training

This section is for anomaly detection, and we studied the steps required to choose, fine-tune, and train a suitable model for anomaly detection. In this, one of the important aspects is model selection. Our proposed model is the deep learning regressor model because it is useful for recognizing complex behaviours and is better than NL techniques. Other than this, it is considered suitable for time series data. Rather than using a classifier, we recommend modelling the issue as a time-series data regression based on a statistical summary window for two reasons: First, a regression can easily model the temporal pattern that signals from ICS sensors and actuators follow. Second, satisfying the desirable accessibility requirement is facilitated using a regression model with an appropriate threshold to provide details regarding the sensor or actuator generating the anomaly. It *****gets time series data in sequence, $(x_0, x_1, x_3, \ldots, x_n)$ for predicting the time series sequence $(x_{n+h}, x_{n+h+1}, x_{n+h+2}, \ldots, x_{n+h+m})$. In this, $m$ is for future predictions, n is window length, and $h$ is the prediction horizon. While $(x_0, x_1, x_3, \ldots, x_n)$ and $x_{n+h+1}, x_{n+h+2}, \ldots, x_{n+h+m})$ are network input and output.

The next goal is to define the ranges of values for each hyperparameter so that they can be fine-tuned during training. Two groups of these hyperparameters can be distinguished. Hyperparameters connected to the input data make up the first group, and those specifically

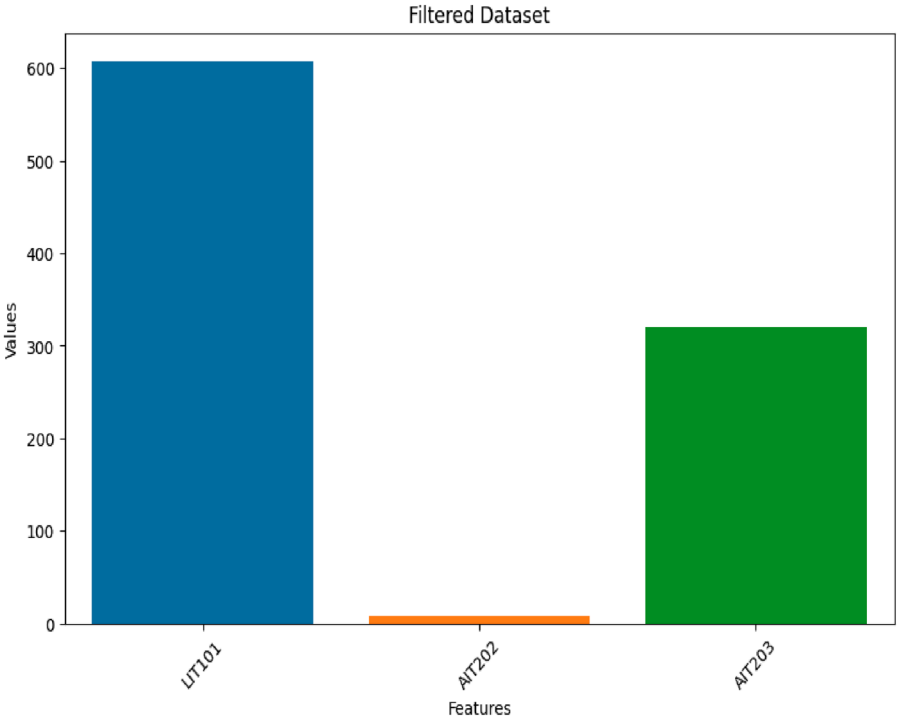



**Figure 8.** Lowest variance features.

related to the deep learning (DL) model make up the second. We suggest the following guidelines to help direct the selection of suitable value ranges for the hyperparameters related to the input data: the *n* window should be bigger for behaviour-capturing; h is for future prediction; and m is the predicted time step required for the time step.

After selecting range values and hyperparameters, the next step is training and fine-tuning the model. There are two well-known approaches for optimal hyperparameter searching: grid search and random search. The grid search approach performs a thorough search by investigating every potential value for the given hyperparameters, while in contrast to this random search strategy, it selects values at random for the designated hyperparameters from a pre-defined list. The selected hyperparameter values are used to train the model for every iteration of the hyperparameter search technique. The training dataset is used in the training process, and the validation dataset is used to assess the performance of the hyperparameters. To be more precise, the validation reconstruction error is used to evaluate the hyperparameter quality, and the configuration with the lowest error is the one that is targeted for optimization.

To identify feasible underfitting or overfitting problems in this task, it is imperative to keep an eye on the training reconstruction error. A high training and validation reconstruction error indicates underfitting, which is the incapacity of the deep learning algorithm to model the data accurately. In another way, overfitting, in which the models place excessive emphasis on memorization of the training data rather than generalization, is indicated when compared to the validation reconstruction error; the training reconstruction error is significantly smaller. Getting more data, using cross-validation, or using regularization techniques to constrain learning are common ways to deal with overfitting. On the other hand, one or more of the following strategies can be used to address underfitting: raising the model's complexity and/or lengthening the training period. These techniques are meant to boost the model's performance and capacity to identify the underlying patterns in the data.

It's crucial to remember that this procedure uses some training datasets as the validation dataset. Therefore, it is inevitable that some important information will be lost during the training phase. The model should be trained using the combined training and validation datasets as the new training dataset after the hyperparameter search has determined the ideal hyperparameter values. By adding the previously omitted data from the validation set, we can do our best to make the most of the available data and possibly enhance the model's performance.

### *3.6.1. Anomaly detection*

In this section, we will cover the threshold adjustment for anomaly detection. We suggest an approach known

as the statistical approach used in previous work (Burgetová et al., 2021; Hao et al., 2023). First, there is error prediction between ground truth and neural networks. Mathematically, it is written as

$$e = |x - \hat{x}| \tag{3}$$

In this Equation (3), $x$ is the ground truth, while $x$ hat is the neural network. This error has been computed for the training dataset. At this stage, standard and mean deviations are calculated.

$$z_e = \frac{v_d - \mu_e}{\omega_e} \tag{4}$$

In this Equation (4) $\mu_e$ is the mean, and $\omega_e$ is the standard deviation, both of which are calculated. Now is the step to find anomaly behaviour in the validation dataset $v_d$. Generally, various approaches are recommended at this stage, like preprocessing the dataset or producing a new $v_d$ that consists of real-valued training and validation datasets. After choosing a method, we use the training dataset's standard deviation and mean deviation to construct a $z$-score for the validation dataset $z_e$.

### 3.7. Improved autoencoder (ImpAE)

AE is an unsupervised approach used to learn and reconstruct its input data. The reconstruction error between the input and output of the AE is used as anomaly detection. To improve the AE performance to avoide from the overfitting and to learn the relevant features quickly, we introducted the "dropout" concept. Key concept to introduce the the dropout layer is to dropout few units and their closed from the network. This step is being done during the training process. It is gained by the hidden layer outcomes from 0.5 to 0. At this level, there is no further contribution of dropped neurons.

### 3.8. Validation process

Various metrics can be used to evaluate the effectiveness of anomaly detection techniques. The following metrics are frequently employed in anomaly detection: Previously, researchers used the F1-score, which is the harmonic means of the precision model (A classification model's ability to recognize only those data points) and recall for the performance of anomaly detection.

- True Positive (TP) generally refers to several correctly identified anomalies or time steps in which the attack was correctly detected.
- True Negative (TN) is the number of normal or non-anomaly instances that were accurately identified.
- False Positive (FP) is the number of normal events mistakenly labelled anomalies (a Type I error).
- False Negative (FN) is the number of anomalies overlooked or mistakenly categorized as typical occurrences (also referred to as a Type II error).
- Precision is gained when the FP rate is divided by TP plus the FN rate.

$$Precision = \frac{TP}{TP + FN} \tag{5}$$

- Recall is also known as sensitivity and is determined when the TP rate is divided by TP plus FN

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

- point-F1 is basically a harmonic means (H.M) of recall/sensitivity and precision scores. multiplication of both of these defines the F1 formula as

$$point - F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

Here, the selected evaluation metric is determined by the problem at hand and the trade-offs between various metrics. Even though F1 score, precision, and recall are crucial metrics in many situations, they might not always be the best option. This study examined a binary classification task where the two classes had equal weight. In these situations, accuracy ensures that no class is given preference over the other by offering a fair assessment of the model's performance on both. Furthermore, when the distribution of classes is balanced, accuracy may be especially important. We recognize that accuracy may not always be the most informative metric when classes are imbalanced. Accuracy is the number of properly predicted occurrences divided by the number of probable instances.

Most importantly, while considering the dataset's unique features, we advise establishing appropriate metrics to validate the model. Accuracy is not the best metric to depend on when working with a dataset that shows a significant imbalance where the number of normal samples is far greater than the number of anomaly samples. Metrics like TP, TN, FP, and FN are better in these situations.

Nevertheless, we suggest including precision, F1-Score, and recall metrics to make comparisons with other studies easier. These metrics provide insightful information about how well the model works and can be used to assess how well it can detect anomalies accurately while reducing false positives and negatives. Evaluating the results' applicability to our particular problem is the second goal. To do this, we need to consider the established metrics and, if practical, contrast our results with

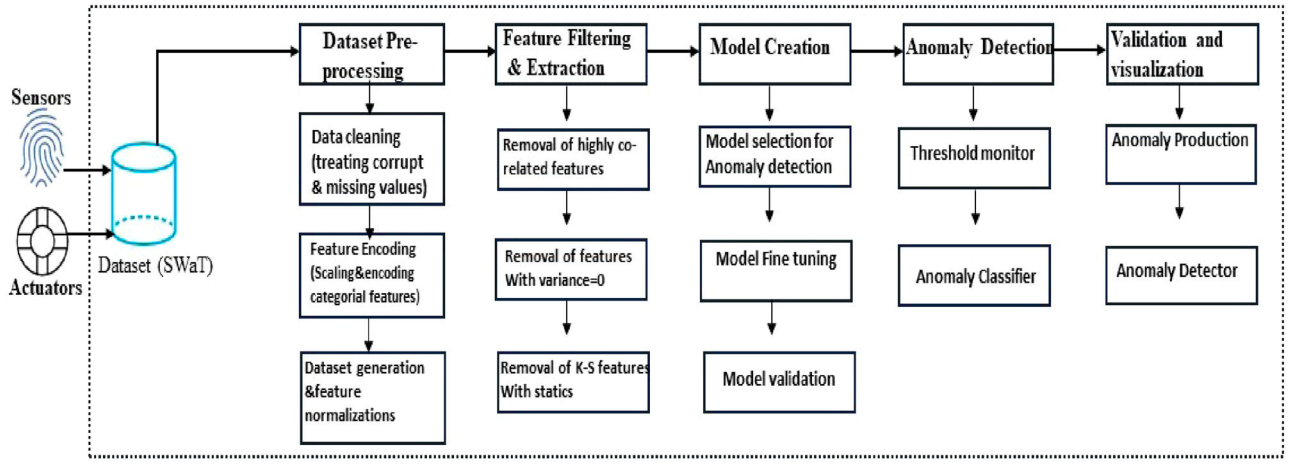



**Figure 9.** Anomaly detection methodology.

those of other research projects that deal with related problems.

Achieving high recall or precision may be emphasized based on our goals. On the one hand, even though it involves some false positives, we should prioritize the recall metric if our main goal is to identify the most significant number of cyberattacks. However, even though it results in some false negatives, we should prioritize the precision metric if our primary goal is to identify cyberattacks with a low false positive rate. Achieving a balance between recall and precision is generally the best option for practice, as the F1-score frequently illustrates. Even though there may be a somewhat higher false positive rate, achieving a high recall rate is still desirable when considering the possible effects of Industrial Control Systems (ICSs) on the real world (Figure 9).

## 4. ICS anomaly detection: applying DL techniques over SWaT dataset

Here, we present the comparison of DL techniques over the SWaT dataset. We study the anomaly detection system for every hyperparameter, as illustrated in Figure 3. The experimental work and the findings of our work are explained in this section.

### *4.1. Experimental configuration*

Here, we explain our experimental configuration, model hyperparameter tuning, and detect hyperparameter tuning. In this experimental setup, for deep learning, we used PyTorch; for preprocessing and evaluation, we used scikit-learn; and for data manipulation, we used pandas and numpy. In our experiment, we used the SWaT dataset as a CSV file. We performed the pre-processing steps like missing values, dropping irrelevant columns, and converting labels to numeric values.

For the experimental configuration, first, for integrity and accuracy, we normalized each feature of the data. Before the model evaluation, the scaling transformation (to fix the data within the required or specific scale) is applied and saved to the attack dataset. For training purposes, 80% of the training dataset is selected to train in deep learning techniques. For the other 20%, fixing it as a benign dataset (whose presence will not affect the correctness of the program) is useful for providing an unbiased score (that ensures impartiality and fairness in analysis by being devoid of systematic or deliberate biases). We employ benign validation loss to avoid overfitting as a signal for early training termination.

In this work, we identify methods that, while applied inconsistently in previous work like early termination in (Erba et al., 2020; Taormina & Galelli, 2018), attack cleaning (Inoue et al., 2017), benign data transforming (Abokifa et al., 2017) and feature selection in Kravchik and Shabtai (2021) and Zizzo et al. (2019) have an impact on the quality and accuracy of results. These techniques (early termination, attack cleaning, benign data transforming, and feature selection) are useful for reproducibility, accuracy, and anomaly detection quality in datasets. Such techniques are mandatory for the comparison of solutions.

Secondly, for model hyperparameter tuning, we used this hyperparameter for the AE technique. We used different numbers of hidden layers, from 1–3 by varying the number of units per layer from 4 to 256 with three kernel sizes (which determine the receptive field, or the input area) and a history length of 200 and 300 timesteps. Activation function sigmoid has been used. The sigmoid function is frequently utilized in the output layer of the decoder to minimize the restored values between 0 and 1. It assists in minimizing the complexity of the AE model by providing a nonlinear transformation and capturing

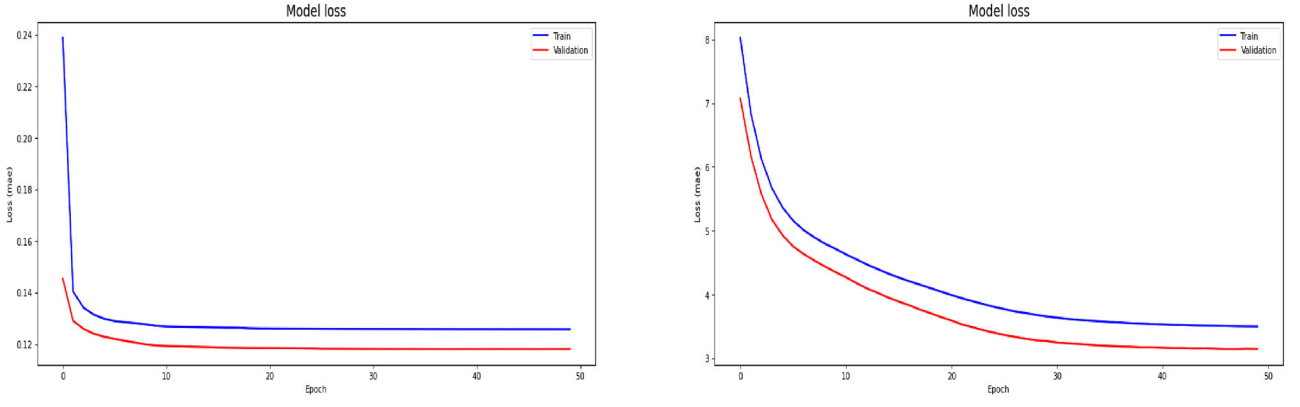
nonlinear relationships in the data. It plays an important role in capturing nonlinear relationships and minimizing the complexity of the AE model, making it useful for anomaly detection. Here, the implementation of the proposed model was done in TensorFlow by using tf.Keras API and Adam Optimizer(used for updating the weights) training with default parameters. This optimizer is popular for neural network training. Each model was trained for up to 50 epochs with a batch size of 512 samples during the training process. Using patience when instructed by tf.keras.callbacks. Patience and call back class with patience $=5$, which is time-saving and effective for termination of training if there is no validation loss for up to five consecutive epochs. Patience was nearly always observed during the first 30 epochs of the training procedure. For this experiment, we used a Graphics Processing Unit (GPU) PhysX NVIDIA GeForceRTX2070, window 11, 64 bit, installed RAM 64.0 GB and processor 12th Gen Intel(R) Core(TM) i9-12900 2.40 GHz based system.

20% of the attack dataset–also known as the attack validation dataset–is used to determine the suitable detection hyperparameters once the model has been trained. To replicate a situation with unidentified attacks, we divided the attack dataset into two continuous sequences before dividing it into testing and validation sections. It means the first 20% of the SWaT dataset has been used for corresponding attack validation. We search parameters over the following ranges using a selected tuning metric to determine the ideal detection hyperparameter values.

## 5. Results and discussion

From the SWaT dataset, we measured performance using F1-score, recall, precision, and accuracy as advised by the first task of the validation methodology step. Our experimental results yielded accuracy for these performance metrics, like the accuracy for the proposed ImpAE, 96% from 93% with baseline AE. The proposed model's high precision value of 0.993 shows that it successfully reduces false positives and has a very low rate of false positives, which makes it dependable for correctly classifying normal instances. Low FPR indicates that the proposed model is sufficient or reliable to differentiate the true positive instances from FP. The model's recall value of 0.673 indicates that it can identify a significant percentage of real anomalies, preventing the omission of important anomalies. With an F1-score of 0.771, the overall performance is well-balanced and shows a positive relationship between recall and precision. These findings demonstrate how well the ImpAE model detects anomalies in the unbalanced SWaT dataset. The attained performance metrics show how well the model detects anomalies and reduces false alarms and missed detections. These results advance our knowledge of anomaly detection in industrial control systems and offer guidance for creating more reliable and accurate models to improve critical infrastructure security. As shown in Table 3, our precision and accuracy results are the best, and the F1-score and recall are better than the average of the works. Training and validation loss comparison of baseline AE and proposed ImpAE have been shown in Figure 10(a,b), respectively. Anomalies in resultant output have been shown in Figure 11

Furthermore, for F1-score, recall, and precision. It shows the number of TP, TN, FP, and FN rates obtained in the validation process. The ImpAE model achieved the highest true positive (78,891), improved than normal AE, and the lowest FP 0, which can be considered a suitable combination in an ICS environment. It is important to note that there are zero false positives (FP) when utilizing the training dataset. This is because the classification procedure uses a threshold calculated from the training/validation dataset. To be more precise, the maximum




(a) Baseline AE (b) ImpAE

**Figure 10.** Training and validation loss of AE and ImpAE models. (a) Baseline AE and (b) ImpAE.

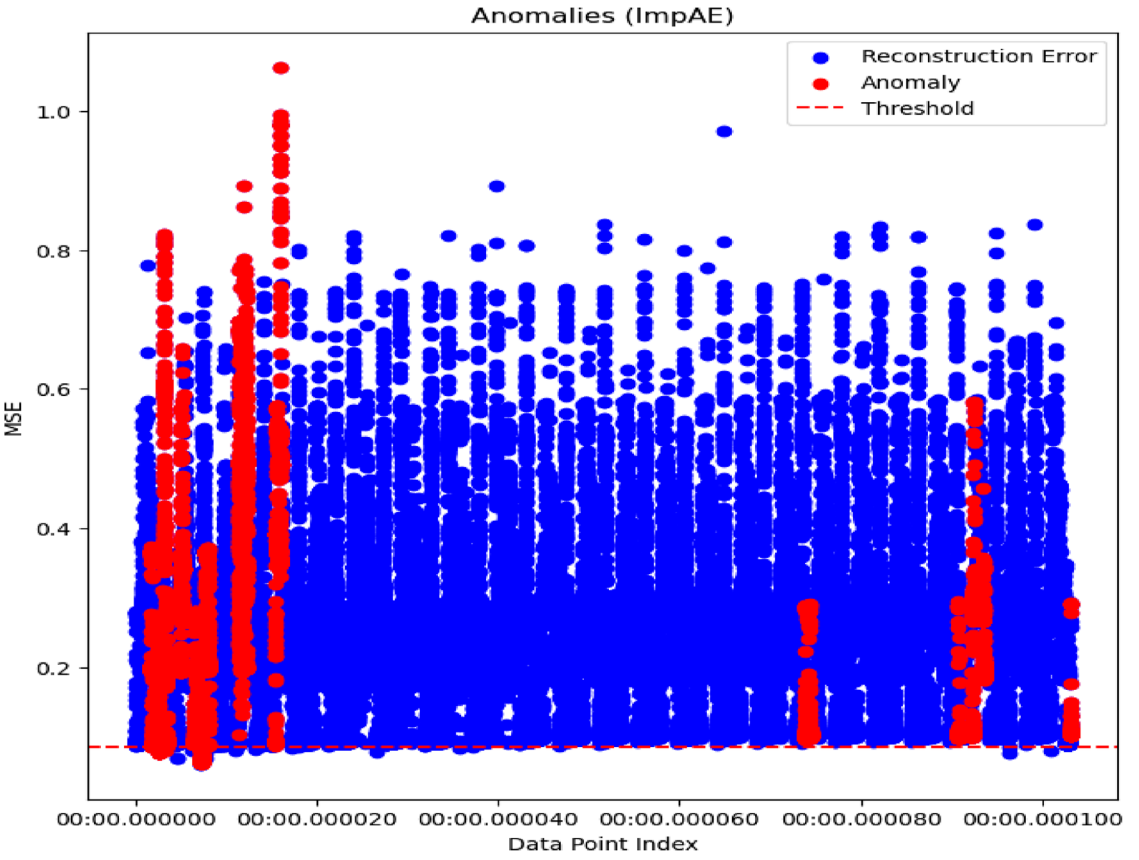



**Figure 11.** Anomalies in ICS (SWaT) dataset.

**Table 3.** Results comparison of various works on SWaT dataset.

| Ref | Model | Precision | Recall | F1-Score | Accuracy |
|---|---|---|---|---|---|
| Kravchik and Shabtai (2021) | 1D CNNs | 0.924 | 0.827 | 0.873 | N/A |
| Inoue et al. (2017) | OCSVM | 0.925 | 0.699 | 0.796 | N/A |
| Elnour et al. (2020) | DIF | 0.935 | 0.835 | **0.882** | N/A |
| Shalyga et al. (2018) | RNN | 0.963 | 0.692 | 0.796 | 95% |
| Shalyga et al. (2018) | CNN | 0.952 | 0.702 | 0.808 | N/A |
| Shalyga et al. (2018) | MLP | 0.967 | 0.696 | 0.812 | N/A |
| Kravchik and Shabtai (2018) | ID CNN | 0.968 | 0.791 | 0.871 | N/A |
| Inoue et al. (2017) | DNN | 0.982 | 0.678 | 0.802 | N/A |
| Li et al. (2019) | LSTM | N/A | N/A | 0.817 | N/A |
| Shalyga et al. (2018) | Genetic algorithms (GA) | 0.967 | 0.702 | 0.812 | 95% |
| Inoue et al. (2017) | One class SVM | 0.9250 | 0.6990 | 0.7928 | N/A |
| Tang et al. (2024) | SAKMR | 0.7833 | 0.8566 | 0.8183 | – |
| Mahmud et al. (2024) | Isolation Forest | N/A | N/A | N/A | 77% |
| Gómez et al. (2023) | LSTM | 0.633 | **0.910** | 0.747 | – |
| **Proposed** | ImpAE | **0.993** | 0.673 | 0.771 | **9**6% |

value attained during the training phase is the chosen threshold. When the point is set to the ultimate value, the model classifies some occurrences as anomalies, which makes it more conservative. Consequently, the training dataset contains no false positives because all occurrences below this threshold are categorized as usual. This method, which minimizes the number of false positives while identifying anomalies accurately, uses an entry taken from the training/validation dataset. It enables the model to create a decision boundary that distinguishes regular occurrences from anomalies using the features and patterns it has learned from the training set. The proposed model shows higher precision in Table 3 by removing false positives from the training dataset, indicating a decreased chance of misunderstanding regular instances as anomalies. This provides additional evidence for the model's dependability and accuracy in identifying abnormalities in the SWaT dataset. While having zero false positives in the training dataset is a positive outcome, it's crucial to remember that the model's generalization ability and performance on unknown data should also be evaluated using cross-validation techniques or

an independent test dataset. This guarantees that the model can accurately identify anomalies in real-world situations and that its performance is not just optimized for the training dataset. Other values gained in our work for research purposes when we were trying to find the best precision with the best method are given in Table 3. recall comparison from other work is given in Table 4. Compared work using the SWaT dataset. We didn't consider those works with detection rate per cyber attack in this Table 4. The objective is anomaly detection or to detect anomalous behaviour in ICS, so we outline recall for such attacks. The proposed work is more useful in detecting cyberattacks in real-world ICS environments. The best recall was 1 for attacks 11, 20, and 22. The second best is for attacks 6 and 28, with a recall of 0.94. The worst is for attack 19, with a recall of 0.69. In the proposed work, we considered a normal behavioural system and considered the attacks given in Table 4. We have already explained the attack effects in Section 3.2.

Table 3 represents the Precision, recall, and F1-Score comparison with other works and DL-used techniques in our results. Here, it is notable that because of insufficient security solutions, specifically in the SWaT environment, other approach outcomes represent overall anomalies available in SWaT. As mentioned above, here we used attacks mentioned in Table 4 only normal behaviour system. In addition, we considered the outcomes of the security breaches detector module. Using the Anomaly Detection Ensemble, which combines various anomaly detectors, will improve these results. With a recall score of 0.925 (0.732 with baseline AE), our work produced the best results. The findings indicated that the precision (0.673) (0.571 with baseline AE) and F1-score (0.771) (0.686 with baseline AE) that cyberattacks that affect the long-term viability of industrial systems can be identified using our technology. Our outcomes show that the proposed work will notify the anomalies ASAP to minimize the impact. Nevertheless, we found that the false positive rate (FPR) was very low when considering that anomaly detection happens in spurts because of the time dependence found in an industrial environment. After removing all of the test dataset's regular traffic to compute the FPR, we discovered that FP occasionally occurred in bursts. Since the system's behaviour does not change much from one instant to the next, each burst can be considered a single alert.

## 6. Discussion and future work

Our proposed method, ImpAE, explained the complete steps for future consideration of cyber security in ICS. We tested the proposed methodology on the SWaT dataset provided by the iTrust Center, Singapore University of Digital Science. Outcomes strongly recommend that ImpAE can be useful for attack detection in an ICS environment. It shows that the proposed technique is useful for preventing physical damage or monitoring loss done by ICS attack effects. ImpAE techniques filled the research gap for attack detection in an ICS environment. Above in Section 2.4, we explained several works using ML and DL techniques, and each work used its own model, and there is no generic model presentation that may lead to wrong in the process. In our best understanding, there are some works like Gómez et al. (2023) and Perales Gómez et al. (2020) aimed at ICS sustainable environment by cyberattack detection. The drawback of these works is that they are limited to cyberattack detection and cannot provide enhanced ML or DL technology in their work.

Our proposed methodology describes a complex ML/DL model in which we did each step in an ICS environment. First, we pre-processed data in Section 3.3, which makes data useful for the proposed model. Here, we suggest treating rough and splitting data and nan values in specific to prevent the removal of specific pieces of information from ICS. Secondly, we did filtering processes studied in Section 3.4, we did three various techniques for filtering rough features (those features which are unable to provide any information), and then thirdly, ICS performed a typical repeated action, and in this, we recommended the extraction of higher order parameters studied in Section 3.5. It has been done in the extraction Section 3.5. The next step is fine-tuning and training, and the basic objective of this step is anomaly detection in the ICS environment studied in Section 3.6. In this step, sensors and actuator information are useful for timestep data. We recommended a based model that is a neural network and is useful for anomaly detection in ICS. In the last step Section 3.8, we did a validation process and recommended particular metrics useful for anomaly detection in the ICS environment.

ImpAE is unsupervised learning, but when it uses large data that combines labelled and unlabelled data, then it becomes in the context of semisupervised learning. During the training process, using labelled data, our proposed model constructs to learn labelled samples accurately, which is useful for anomaly detection. The proposed model reconstructs normal samples better than anomalous samples. In our work, we highlighted the importance of pre-processing and extraction. In our proposed work, this pre-processing allows us to ignore 100000 datasets during the warm-up process. Also, in the filtering process, we removed rough readings.

Compare this to previous research on anomaly detection studied in Section 2.4, previous work methodologies

**Table 4.** Comparison of various works on SWaT dataset (Recall Comparison).

| Attack | ID-CNN (Kravchik & Shabtai, 2018) | RNN (Shalyga et al., 2018) | DNN (Inoue et al., 2017) | OCSVM (Inoue et al., 2017) | DIF (Elnour et al., 2020) | LSTM (Gómez et al., 2023) | CNN (Perales Gómez et al., 2020) | Proposed |
|---|---|---|---|---|---|---|---|---|
| 6 | 0.90 | 0.72 | 0.95 | 0.72 | 1 | 0.92 | 0 | 0.94 |
| 11 | 1 | 0.98 | 0.99 | 1 | 1 | 0.96 | 1 | 1 |
| 19 | 0 | 0.12 | 0.97 | 0.13 | 0.45 | 0.64 | 0 | 0.69 |
| 20 | 1 | 0.85 | 0 | 0.85 | 0.45 | 1 | 0 | 1 |
| 22 | 1 | 0.99 | 0.98 | 1 | 1 | 0.97 | 1 | 1 |
| 24 | 0.17 | 0 | 0.92 | 0 | 0.34 | 0.78 | 0.53 | 0.85 |
| 28 | 1 | 0.94 | 0.03 | 0.94 | 1 | 0.93 | 0.97 | 0.94 |
| 38 | 0.86 | 0.93 | 0.77 | 0.93 | 1 | 0.69 | 0.01 | 0.72 |
| 40 | 1 | 0.92 | 0.78 | 0.93 | 1 | 0.74 | 1 | 0.77 |

are unclear and noticed with errors. The general error is to ignore the warm-up process in the dataset.

Authors in previous works didn't focus on the ICS nature; for example, ICS repeated nature. Together with the earlier methodological flaws, the absence of several steps and the use of poor approaches suggest that the results reported in those publications were skewed, enhancing the results in most instances. Furthermore, these methodological mistakes highlight how urgently an ICS-focussed standard methodology is needed. We proved that the proposed model achieved better and more useful results, as shown in Tables 3 and 4. In the future, our work will require two datasets: when the ICS environment is normally working and when an ICS is working under an attack scenario. Currently, limited datasets are available for research purposes to implement the proposed model on different datasets. Because of this, there is no generic model for the security of the ICS environment. In the future, we would like to implement our model in different industrial sectors like the power sector, energy sector, critical infrastructure, and many more with the availability of the dataset. Because of the unavailability of the datasets for various sectors, developing a generic model for researchers is a key challenge in ICS security. Current research in ICS security has suffered from high false positive rates, computational complexity, and accuracy; in the future, we need to develop a generic model for a secure ICS environment with the lowest possible risk.

## 7. Conclusion

In this paper, we presented ImpAE for anomaly detection in ICS. This model consists of five steps, as studied in Section 3. These steps are data preprocessing Section 3.3, Filtring Section 3.4, Extraction Section 3.5, fine-tuning and training Section 3.6 and validation process Section 3.8. Furthermore, a few procedures were conducted to show that the ImpAE framework is feasible. Specifically, we demonstrated how the various components were implemented and how components in a SWaT were used to identify security abnormalities brought on by cyberattacks. ITrust Center, Singapore University of Digital Science, Singapore, provided the SWaT dataset. Data consists of a fully operating water testbed plant for 11 days, seven days of data under normal operation, and four days of data under attacks. The most balanced results were obtained by our implementation, which also identified every cyberattack that resulted in security irregularities. Precision 0.993 from 0.735, accuracy 96%, and recall and F1-Score were beyond the average of compared literature work. Regarding the attack detection consideration, the proposed model could detect the maximum attack, for attacks 11, 20, and 22 recall was 1 while for other attacks close to 1. In future consideration, we plan to do further experiments and check validations in other datasets like WADI, smart industry, oil & gas, and BATADAL. In addition to this, we plan to work on a sustainable ICS environment.

## Acknowledgments

Conceptualization, M.M.A. and A.T.; Paper flow, A.N. and Liyang C.D.S; software, M.M.A.; validation, R.A.A.H.M.A, Liyang C.D.S andA.T.; formal analysis Liyang C.D.S; investigation R.A.A.H, M.A., A.N., and A.T.; data curation, M.M.A.; writing–original draft preparation, M.M.A.; writing–review and editing, M.M.A., A.T., Liyang C.D.S, R.A.A.H.M.A. and A.N.; supervision, A.T., Liyang C.D.S, and R.A.A.H; project administration, A.T,; funding acquisition, A.T. All authors have read and agreed to the published version of the manuscript.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Funding

Funding support from Universiti Brunei Darussalam, Brunei.

## Data availability

The iTrust centre, Singapore University of Digital Science, Singapore, provided data. https://shorturl.at/wY179

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